PSY 590 : Special Topics in Psychological Science

(1-3 credits) This course provides an opportunity for graduate students to learn about a special topic within Psychological Science. This course may be repeated for different topics.

Department

Psychology

Mathematical Sciences

BCB 504/MA 584 : Statistical Methods in Genetics and Bioinformatics

This course provides students with knowledge and understanding of the applications of statistics in modern genetics and bioinformatics. The course generally covers population genetics, genetic epidemiology, and statistical models in bioinformatics. Specific topics include meiosis modeling, stochastic models for recombination, linkage and association studies (parametric vs. nonparametric models, family-based vs. population-based models) for mapping genes of qualitative and quantitative traits, gene expression data analysis, DNA and protein sequence analysis, and molecular evolution. Statistical approaches include log-likelihood ratio tests, score tests, generalized linear models, EM algorithm, Markov chain Monte Carlo, hidden Markov model, and classification and regression trees. Students may not receive credit for both BCB 4004 and BCB 504.

Department

Bioinformatics and Computational Biology

Mathematical Sciences

Credits 3.0

Prerequisites

knowledge of probability and statistics at the undergraduate level

CS 522/MA 510 : Numerical Methods

See MA 510 course description.

Department

Computer Science

Mathematical Sciences

Credits 3.0

DS 502/MA 543 : Statistical Methods for Machine Learning

Statistical Methods for Machine Learning surveys the statistical methods most useful in machine learning applications. Topics covered include predictive modeling methods, including multiple linear regression, and time series, data dimension reduction, discrimination and classification methods, clustering methods, and committee methods. Students will implement these methods using statistical software.

Department

Data Science

Mathematical Sciences

Credits 3.0

Prerequisites

DS 5002/MA 517, Statistics at the level of MA 2611 and MA 2612 and linear algebra at the level of MA 2071.

DS 5002/MA 517 : Introductory Statistical Methods for Machine Learning

The foci of this class are the essential statistics and linear algebra skills required for Data Science students. The class builds the foundation for theoretical and computational abilities of the students to analyze high dimensional data sets. Topics covered include Bayes' theorem, the central limit theorem, hypothesis testing, linear equations, linear transformations, matrix algebra, eigenvalues and eigenvectors, and sampling techniques, including Bootstrap and Markov chain Monte Carlo. Students will use these techniques while engaging in hands-on projects with real data.

Department
Data Science
Mathematical Sciences
Credits 3.0
Prerequisites
Some knowledge of integral and differential calculus is recommended.

MA 500 : Basic Real Analysis

This course covers basic set theory, topology of Rn, continuous functions, uniform convergence, compactness, infinite series, theory of differentiation and integration. Other topics covered may include the inverse and implicit function theorems and Riemann-Stieltjes integration. Students may not count both MA 3831 and MA 300 toward their undergraduate degree requirements.

Department
Mathematical Sciences
Credits 3.0

MA 501 : Engineering Mathematics

This course develops mathematical techniques used in the engineering disciplines. Preliminary concepts will be reviewed as necessary, including vector spaces, matrices and eigenvalues. The principal topics covered will include vector calculus, Fourier transforms, fast Fourier transforms and Laplace transformations. Applications of these techniques for the solution of boundary value and initial value problems will be given. The problems treated and solved in this course are typical of those seen in applications and include problems of heat conduction, mechanical vibrations and wave propagation.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
A knowledge of ordinary differential equations, linear algebra and multivariable calculus is assumed

MA 502 : Linear Algebra

This course provides an introduction to the theory and methods of applicable linear algebra. The goal is to bring out the fundamental concepts and techniques that underlie and unify the many ways in which linear algebra is used in applications. The course is suitable for students in mathematics and other disciplines who wish to obtain deeper insights into this very important subject than are normally offered in undergraduate courses. It is also intended to provide a foundation for further study in subjects such as numerical linear algebra and functional analysis.

Department
Mathematical Sciences
Credits 3.0

MA 503 : Lebesgue Measure and Integration

This course begins with a review of topics normally covered in undergraduate analysis courses: open, closed and compact sets; liminf and limsup; continuity and uniform convergence. Next the course covers Lebesgue measure in Rn including the Cantor set, the concept of a sigma-algebra, the construction of a nonmeasurable set, measurable functions, semicontinuity, Egorovs and Lusin's theorems, and convergence in measure. Next we cover Lebesgue integration, integral convergence theorems (monotone and dominated), Tchebyshev's inequality and Tonelli's and Fubini's theorems. Finally Lp spaces are introduced with emphasis on L2 as a Hilbert space. Other related topics will be covered at the instructor's discretion.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
Basic knowledge of undergraduate analysis is assumed

MA 504 : Functional Analysis

This course will give a comprehensive presentation of fundamental concepts and theorems in Banach and Hilbert spaces. Whenever possible, the theory will be illustrated by examples in Lebesgue spaces. Topics include: The Hahn-Banach theorems, the Uniform Boundedness principle (Banach-Steinhaus Theorem), the Open Mapping and Closed Graph theorems, and weak topologies and convergence. Additional topics will be covered at the instructor's discretion.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
MA 503 or equivalent

MA 505 : Complex Analysis

This course will provide a rigorous and thorough-treatment of the theory of functions of one complex variable. The topics to be covered include complex numbers, complex differentiation, the Cauchy-Riemann equations, analytic functions, Cauchy's theorem, complex integration, the Cauchy integral formula, Liouville's theorem, the Gauss mean value theorem, the maximum modulus theorem, Rouche's theorem, the Poisson integral formula, Taylor-Laurent expansions, singularity theory, conformal mapping with applications, analytic continuation, Schwarz's reflection principle and elliptic functions.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of undergraduate analysis

MA 508 : Mathematical Modeling

This course introduces mathematical modelbuilding using dimensional analysis, perturbation theory and variational principles. Models are selected from the natural and social sciences according to the interests of the instructor and students. Examples are: planetary orbits, spring-mass systems, fluid flow, isomers in organic chemistry, biological competition, biochemical kinetics and physiological flow. Computer simulation of these models will also be considered.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of ordinary differential equations and of analysis at the level of MA 501 is assumed

MA 509 : Stochastic Modeling

This course gives students a background in the theory and methods of probability, stochastic processes and statistics for applications. The course begins with a brief review of basic probability, discrete and continuous random variables, expectations, conditional probability and basic statistical inference. Topics covered in greater depth include generating functions, limit theorems, basic stochastic processes, discrete and continuous time Markov chains, and basic queuing theory including M/M/1 and M/G/l queues. This course is offered by special arrangement only, based on expressed student interest.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of basic probability at the level of MA 2631 and statistics at the level of MA 2612 is assumed.

MA 510/CS 522 : Numerical Methods

This course provides an introduction to a broad range of modern numerical techniques that are widely used in computational mathematics, science, and engineering. It is suitable for both mathematics majors and students from other departments. It covers introductory-level material for subjects treated in greater depth in MA 512 and MA 514 and also topics not addressed in either of those courses. Subject areas include numerical methods for systems of linear and nonlinear equations, interpolation and approximation, differentiation and integration, and differential equations. Specific topics include basic direct and iterative methods for linear systems; classical rootfinding methods; Newton's method and related methods for nonlinear systems; fixed-point iteration; polynomial, piecewise polynomial, and spline interpolation methods; least-squares approximation; orthogonal functions and approximation; basic techniques for numerical differentiation; numerical integration, including adaptive quadrature; and methods for initial-value problems for ordinary differential equations. Additional topics may be included at the instructor's discretion as time permits. Both theory and practice are examined. Error estimates, rates of convergence, and the consequences of finite precision arithmetic are also discussed. Topics from linear algebra and elementary functional analysis will be introduced as needed. These may include norms and inner products, orthogonality and orthogonalization, operators and projections, and the concept of a function space.

Department

Mathematical Sciences

Computer Science

Credits 3.0

Prerequisites

knowledge of undergraduate linear algebra and differential equations is assumed, as is familiarity with MATLAB or a higher-level programming language

MA 511 : Applied Statistics for Engineers and Scientists

This course is an introduction to statistics for graduate students in engineering and the sciences. Topics covered include basic data analysis, issues in the design of studies, an introduction to probability, point and interval estimation and hypothesis testing for means and proportions from one and two samples, simple and multiple regression, analysis of one and two-way tables, one-way analysis of variance. As time permits, additional topics, such as distribution-free methods and the design and analysis of factorial studies will be considered.

Department

Mathematical Sciences

Credits 3.0

Prerequisites

Integral and differential calculus

MA 512 : Numerical Differential Equations

This course begins where MA 510 ends in the study of the theory and practice of the numerical solution of differential equations. Central topics include a review of initial value problems, including Euler's method, Runge-Kutta methods, multi-step methods, implicit methods and predictor-corrector methods; the solution of two-point boundary value problems by shooting methods and by the discretization of the original problem to form systems of nonlinear equations; numerical stability; existence and uniqueness of solutions; and an introduction to the solution of partial differential equations by finite differences. Other topics might include finite element or boundary element methods, Galerkin methods, collocation, or variational methods.

Department

Mathematical Sciences

Credits 3.0

Prerequisites

graduate or undergraduate numerical analysis. Knowledge of a higher-level programming language is assumed

MA 514 : Numerical Linear Algebra

This course provides students with the skills necessary to develop, analyze and implement computational methods in linear algebra. The central topics include vector and matrix algebra, vector and matrix norms, the singular value decomposition, the LU and QR decompositions, Householder transformations and Givens rotations, and iterative methods for solving linear systems including Jacobi, Gauss-Seidel, SOR and conjugate gradient methods; and eigenvalue problems. Applications to such problem areas as least squares and optimization will be discussed. Other topics might include: special linear systems, such as symmetric, positive definite, banded or sparse systems; preconditioning; the Cholesky decomposition; sparse tableau and other least-square methods; or algorithms for parallel architectures.

Department

Mathematical Sciences

Credits 3.0

Prerequisites

basic knowledge of linear algebra or equivalent background. Knowledge of a higher-level programming language is assumed

MA 517/DS 5002 : Introductory Statistical Methods for Machine Learning

The foci of this class are the essential statistics and linear algebra skills required for Data Science students. The class builds the foundation for theoretical and computational abilities of the students to analyze high dimensional data sets. Topics covered include Bayes' theorem, the central limit theorem, hypothesis testing, linear equations, linear transformations, matrix algebra, eigenvalues and eigenvectors, and sampling techniques, including Bootstrap and Markov chain Monte Carlo. Students will use these techniques while engaging in hands-on projects with real data.

Department

Mathematical Sciences

Data Science

Credits 3.0

Prerequisites

Some knowledge of integral and differential calculus is recommended.

MA 520 : Fourier Transforms and Distributions

The course will cover L1, L2, L°° and basic facts from Hilbert space theory (Hilbert basis, projection theorems, Riesz theory). The first part of the course will introduce Fourier series: the L2 theory, the C°° theory: rate of convergence, Fourier series of real analytic functions, application to the trapezoidal rule, Fourier transforms in L1, Fourier integrals of Gaussians, the Schwartz class S, Fourier transforms and derivatives, translations, convolution, Fourier transforms in L2, and characteristic functions of probability distribution functions. The second part of the course will cover tempered distributions and applications to partial differential equations. Other related topics will be covered at the instructor's discretion.

Department

Mathematical Sciences

Credits 3.0

Prerequisite Courses

MA 503

MA 521 : Partial Differential Equations

This course considers a variety of material in partial differential equations (PDE). Topics covered will be chosen from the following: classical linear elliptic, parabolic and hyperbolic equations and systems, characteristics, fundamental/Green's solutions, potential theory, the Fredholm alternative, maximum principles, Cauchy problems, Dirichlet/Neumann/Robin problems, weak solutions and variational methods, viscosity solutions, nonlinear equations and systems, wave propagation, free and moving boundary problems, homogenization. Other topics may also be covered.

Department

Mathematical Sciences

Credits 3.0

Prerequisites

MA 503 or equivalent

MA 522 : Hilbert Spaces and Applications to PDE

The course covers Hilbert space theory with special emphasis on applications to linear ODs and PDEs. Topics include spectral theory for linear operators in n-dimensional and infinite dimensional Hilbert spaces, spectral theory for symmetric compact operatos, linear and bilinear forms, Riesz and Lax-Milgram theorems, weak derivatives, Sobolev spaces H1, H2, Rellich compactness theorem, weak and classical solutions for Dirichlet and Neumann problems in one variable and in Rn, Dirichlet variational principle, eigenvalues and eigenvectors. Other related topics will be covered at the instructor's discretion.

Department
Mathematical Sciences
Credits 3.0
Prerequisite Courses
MA 503

MA 524 : Convex Analysis and Optimization

This course covers topics in functional analysis that are critical to the study of convex optimization problems. The first part of the course will include the minimization theory for quadratic and convex functionals on convex sets and cones, the Legendre-Fenchel duality, variational inequalities and complementarity systems. The second part will include optimal stopping time problems in deterministic control, value functions and Hamilton-Jacobi inequalities and linear and quadratic programming, duality and Kuhn-Tucker multipliers. Other related topics will be covered at the instructor's discretion.

Department
Mathematical Sciences
Credits 3.0
Prerequisite Courses
MA 503

MA 528 : Measure Theoretic Probability Theory

This course is designed to give graduate students interested in financial mathematics and stochastic analysis the necessary background in measure-theoretic probability and provide a theoretical foundation for Ph.D. students with research interests in analysis and mathematical statistics. Besides classical topics such as the axiomatic foundations of probability, conditional probabif ities and independence, random variables and their distributions, and limit theorems, this course focuses on concepts crucial for the understanding of stochastic processes and quantitative finance: conditional expectations, filtrations and martingales as well as change of measure techniques and the Radon-Nikodym theorem. A wide range of illustrative examples from a topic chosen by the instructor's discretion (e.g financial mathematics, signal processing, actuarial mathematics) will be presented.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
MA 500 Basic Real Analysis or equivalent

MA 529 : Stochastic Processes

This course is designed to introduce students to continuous-time stochastic processes. Stochastic processes play a central role in a wide range of applications from signal processing to generative A.I. to finance and offer an alternative novel viewpoint to several areas of mathematical analysis, such as partial differential equations and potential theory. The first part of this course will cover the theory of stochastic processes, including martingales, Brownian motion and diffusions, stochastic differential equations, stochastic (Ito) calculus, and Markov Chains. The second part of the course will cover applications chosen by the instructor, such as simulation of stochastic processes, randomized algorithms and applications, stochastic optimization, spatial-temporal statistics, nonlinear filtering, applications to deep learning and generative A.I., or applications in finance. Students are encouraged to ask the instructor for a list of the covered applications.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
Calculus-based probability, statistics, linear algebra, experience with upper-level mathematics or mathematically oriented courses from different disciplines, such as computer science, data science, or physics.

MA 530 : Discrete Mathematics

This course provides the student of mathematics or computer science with an overview of discrete structures and their applications, as well as the basic methods and proof techniques in combinatorics. Topics covered include sets, relations, posets, enumeration, graphs, digraphs, monoids, groups, discrete probability theory and propositional calculus.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
college math at least through calculus. Experience with recursive programming is helpful, but not required

MA 533 : Discrete Mathematics II

This course is designed to provide an in-depth study of some topics in combinatorial mathematics and discrete optimization. Topics may vary from year to year. Topics covered include, as time permits, partially ordered sets, lattices, matroids, matching theory, Ramsey theory, discrete programming problems, computational complexity of algorithms, branch and bound methods.

Department
Mathematical Sciences
Credits 3.0

MA 535 : Algebra

Fundamentals of group theory: homomorphisms and the isomorphism theorems, finite groups, structure of finitely generated Abelian groups. Structure of rings: homomorphisms, ideals, factor rings and the isomorphism theorems, integral domains, factorization. Field theory: extension fields, finite fields, theory of equations. Selected topics from: Galois theory, Sylow theory, Jordan-Holder theory, Polya theory, group presentations, basic representation theory and group characters, modules. Applications chosen from mathematical physics, Grobner bases, symmetry, cryptography, error-correcting codes, number theory.

Department
Mathematical Sciences
Credits 3.0

MA 540/4631 : Probability and Mathematical Statistics I

Intended for advanced undergraduates and beginning graduate students in the mathematical sciences, and for others intending to pursue the mathematical study of probability and statistics. Topics covered include axiomatic foundations, the calculus of probability, conditional probability and independence, Bayes' Theorem, random variables, discrete and continuous distributions, joint, marginal and conditional distributions, covariance and correlation, expectation, generating functions, exponential families, transformations of random variables, types of convergence, laws of large numbers the Central Limit Theorem, Taylor series expansion, the delta method.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of basic probability at the level of MA 2631 and of advanced calculus at the level of MA 3831/3832 is assumed

MA 541/4632 : Probability and Mathematical Statistics II

This course is designed to provide background in principles of statistics. Topics covered include estimation criteria: method of moments, maximum likelihood, least squares, Bayes, point and interval estimation, Fisher's information, Cramer-Rao lower bound, sufficiency, unbiasedness, and completeness, Rao-Blackwell Theorem, efficiency, consistency, interval estimation pivotal quantities, Neyman-Person Lemma, uniformly most powerful tests, unbiased, invariant and similar tests, likelihood ratio tests, convex loss functions, risk functions, admissibility and minimaxity, Bayes decision rules.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of the material in MA 340 is assumed

MA 542 : Regression Analysis

Regression analysis is a statistical tool that utilizes the relation between a response variable and one or more predictor variables for the purposes of description, prediction and/or control. Successful use of regression analysis requires an appreciation of both the theory and the practical problems that often arise when the technique is employed with real-world data. Topics covered include the theory and application of the general linear regression model, model fitting, estimation and prediction, hypothesis testing, the analysis of variance and related distribution theory, model diagnostics and remedial measures, model building and validation, and generalizations such as logistic response models and Poisson regression. Additional topics may be covered as time permits. Application of theory to real-world problems will be emphasized using statistical computer packages.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of probability and statistics at the level of MA 311 and of matrix algebra is assumed

MA 543/DS 502 : Statistical Methods for Machine Learning

Statistical Methods for Machine Learning surveys the statistical methods most useful in machine learning applications. Topics covered include predictive modeling methods, including multiple linear regression, and time series, data dimension reduction, discrimination and classification methods, clustering methods, and committee methods. Students will implement these methods using statistical software. Prerequisites: DS 5002/ MA 517, Statistics at the level of MA 2611 and MA 2612 and linear algebra at the level of MA 2071.

Department
Mathematical Sciences
Data Science
Credits 3.0
Prerequisites
DS 5002/MA 517, Statistics at the level of MA 2611 and MA 2612 and linear algebra at the level of MA 2071.

MA 544/SS 510 : Principles of Epidemiology

Epidemiology studies the pattern of disease in populations to describe and identify distributions of diseases and opportunities for intervention. This course serves as a cornerstone for the quantitative aspects of global health and focuses on the distribution and determinants of health in human populations and communities. The goal is to provide a scientific foundation for evaluating both risk factors and interventions to improve health in a population through a strong quantitative analysis of causation, problem-solving, and analytic reasoning. The study of epidemiology evaluates the multifactorial etiology and pathophysiology of noncommunicable and infectious diseases and contributes to public health practice and policy. Specific topics include biomedical study design (i.e., experiment, cohort, case-control, cross sectional, ecological), appropriate measures of disease burden and association (i.e., prevalence, cumulative incidence, rate ratio, odds ratio), and considerations for efficacy and precision (i.e., selection bias, confounding, effect modification, measurement error, and random variation). The course also provides a framework for understanding and evaluating biomedical research publications, causal inference, and basic infectious disease modeling.

Department
Mathematical Sciences
General Social Science
Category
Category I
Credits 3.0

MA 546 : Design and Analysis of Experiments

Controlled experiments—studies in which treatments are assigned to observational units— are the gold standard of scientific investigation. The goal of the statistical design and analysis of experiments is to (1) identify the factors which most affect a given process or phenomenon; (2) identify the ways in which these factors affect the process or phenomenon, both individually and in combination; (3) accomplish goals 1 and 2 with minimum cost and maximum efficiency while maintaining the validity of the results. Topics covered in this course include the design, implementation and analysis of completely randomized complete block, nested, split plot, Latin square and repeated measures designs. Emphasis will be on the application of the theory to real data using statistical computer packages.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of basic probability and statistics at the level of MA 511 is assumed

MA 547 : Design and Analysis of Observational and Sampling Studies

Like controlled experiments, observational studies seek to establish cause-effect relationships, but unlike controlled experiments, they lack the ability to assign treatments to observational units. Sampling studies, such as sample surveys, seek to characterize aspects of populations by obtaining and analyzing samples from those populations. Topics from observational studies include: prospective and retrospective studies; overt and hidden bias; adjustments by stratification and matching. Topics from sampling studies include: simple random sampling and associated estimates for means, totals, and proportions; estimates for subpopulations; unequal probability sampling; ratio and regression estimation; stratified, cluster, systematic, multistage, double sampling designs, and, time permitting, topics such as model-based sampling, spatial and adaptive sampling.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of basic probability and statistics, at the level of MA 511 is assumed

MA 548 : Quality Control

This course provides the student with the basic statistical tools needed to evaluate the quality of products and processes. Topics covered include the philosophy and implementation of continuous quality improvement methods, Shewhart control charts for variables and attributes, EWMA and Cusum control charts, process capability analysis, factorial and fractional factorial experiments for process design and improvement, and response surface methods for process optimization. Additional topics will be covered as time permits. Special emphasis will be placed on realistic applications of the theory using statistical computer packages.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of basic probability and statistic, at the level of MA 511 is assumed

MA 549 : Analysis of Lifetime Data

Lifetime data occurs frequently in engineering, where it is known as reliability or failure time data, and in the biomedical sciences, where it is known as survival data. This course covers the basic methods for analyzing such data. Topics include: probability models for lifetime data, censoring, graphical methods of model selection and analysis, parametric and distribution-free inference, parametric and distribution-free regression methods. As time permits, additional topics such as frailty models and accelerated life models will be considered. Special emphasis will be placed on realistic applications of the theory using statistical computer packages.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of basic probability and statistics at the level of MA 511 is assumed

MA 550 : Time Series Analysis

Time series are collections of observations made sequentially in time. Examples of this type of data abound in many fields ranging from finance to engineering. Special techniques are called for in order to analyze and model these data. This course introduces the student to time and frequency domain techniques, including topics such as autocorrelation, spectral analysis, and ARMA and ARIMA models, Box-Jenkins methodology, fitting, forecasting, and seasonal adjustments. Time permitting, additional topics will be chosen from: Kalman filter, smoothing techniques, Holt-Winters procedures, FARIMA and GARCH models, and joint time-frequency methods such as wavelets. The emphasis will be in application to real data situations using statistical computer packages.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of MA 511 is assumed. Knowledge of MA 541 is also assumed, but may be taken concurrently

MA 551 : Computational Statistics

Computational statistics is an essential component of modern statistics that often requires efficient algorithms and programing strategies for statistical learning and data analysis. This course will introduce principles and techniques of statistical computing and data management necessary for computationally intensive statistical analysis especially for big data. Topics covered include management of large data (data structure, data query), parallelized data analyses, stochastic simulations (Monte Carlo methods, permutation-based inference), numerical optimization in statistical inference (deterministic and stochastic convex analysis, EM algorithm, etc.), randomization methods (bootstrap methods), etc. Students will use these techniques while engaging in hands-on projects with real data. Students who have taken the MA590 version of this course cannot also earn credit for MA 551.

Department
Mathematical Sciences
Prerequisites
No previous programming knowledge/experience is assumed. Some knowledge of probability and statistics, or MA511 equivalent is recommended.

MA 552 : Distribution-Free and Robust Statistical Methods

Distribution-free statistical methods relax the usual distributional modeling assumptions of classical statistical methods. Robust methods are statistical procedures that are relatively insensitive to departures from typical assumptions, while retaining the expected behavior when assumptions are satisfied. Topics covered include, time permitting, order statistics and ranks; classical distribution-free tests such as the sign, Wilcoxon signed rank, and Wilcoxon rank sum tests, and associated point estimators and confidence intervals; tests pertaining to one and two-way layouts; the Kolmogorov-Smirnov test; permutation methods; bootstrap and Monte Carlo methods; M, L, and R estimators, regression, kernel density estimation and other smoothing methods. Comparisons will be made to standard parametric methods.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of MA 541 is assumed, but may be taken concurrently

MA 554 : Applied Multivariate Analysis

This course is an introduction to statistical methods for analyzing multivariate data. Topics covered are multivariate sampling distributions, tests and estimation of multivariate normal parameters, multivariate ANOVA, regression, discriminant analysis, cluster analysis, factor analysis and principal components. Additional topics will be covered as time permits. Students will be required to analyze real data using one of the standard packages available.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of MA 541 is assumed, but may be taken concurrently. Knowledge of matrix algebra is assumed

MA 556 : Applied Bayesian Statistics

Bayesian statistics makes use of an inferential process that models data summarizing the results in terms of probability distributions for the model parameters. A key feature is that in the Bayesian approach, past information can be updated with new data in an elegant way in order to aid in decision making. Topics included in the courses: statistical decision theory, the Bayesian inferential framework (model specification, model fitting and model checking); computational methods for posterior simulation integration; regression models, hierarchical models, and ANOVA; time permitting, additional topics will include generalized linear models, multivariate models, missing data problems, and time series analysis.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
knowledge of MA 541 is assumed

MA 557 : Graduate Seminar in Applied Mathematics

This seminar introduces students to modern issues in Applied Mathematics. In the seminar, students and faculty will read and discuss survey and research papers, make and attend presentations, and participate in brainstorming sessions toward the solution of advanced mathematical problems.

Department
Mathematical Sciences
Credits 0.0

MA 559 : Statistics Graduate Seminar

This seminar introduces students to issues and trends in modern statistics. In the seminar, students and faculty will read and discuss survey and research papers, make and attend presentations, and participate in brainstorming sessions toward the solution of advanced statistical problems.

Department
Mathematical Sciences
Credits 1.0

MA 560 : Graduate Seminar

Designed to introduce graduate students to study of original papers and afford them opportunity to give account of their work by talks in the seminar.

Department
Mathematical Sciences
Credits 0.0

MA 562 A and B. : Professional Master's Seminar

This seminar will introduce professional masters students to topics related to general writing, presentation, group communication and interviewing skills, and will provide the foundations to successful cooperation within interdisciplinary team environments. All full-time students will be required to take both components A and B of the seminar during their professional masters studies.

Department
Mathematical Sciences
Credits 0.0

MA 571 : Financial Mathematics I

This course provides an introduction to many of the central concepts in mathematical finance. The focus of the course is on arbitrage-based pricing of derivative securities. Topics include stochastic calculus, securities markets, arbitrage-based pricing of options and their uses for hedging and risk management, forward and futures contracts, European options, American options, exotic options, binomial stock price models, the Black-Scholes-Merton partial differential equation, risk-neutral option pricing, the fundamental theorems of asset pricing, sensitivity measures ("Greeks"), and Merton's credit risk model.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
MA 540, which can be taken concurrently

MA 572 : Financial Mathematics II

The course is devoted to the mathematics of fixed income securities and to the financial instruments and methods used to manage interest rate risk. The first topics covered are the term-structure of interest rates, bonds, futures, interest rate swaps and their uses as investment or hedging tools and in asset-liability management. The second part of the course is devoted to dynamic term-structure models, including risk-neutral interest rate trees, the Heath-Jarrow-Morton model, Libor market models, and forward measures. Applications of these models are also covered, including the pricing of non-linear interest rate derivatives such as caps, floors, collars, swaptions and the dynamic hedging of interest rate risk. The course concludes with the coverage of mortgage-backed and asset-backed securities.

Department
Mathematical Sciences
Credits 3.0
Prerequisite Courses
MA 571

MA 573 : Computational Methods of Financial Mathematics

Most realistic quantitative finance models are too complex to allow explicit analytic solutions and are solved by numerical computational methods. The first part of the course covers the application of finite difference methods to the partial differential equations arising in option pricing and model calibration. Topics included are explicit, implicit and Crank-Nicholson finite difference schemes for fixed and free boundary value problems, their convergence and stability. The second part of the course focuses on modern advancements in financial computational methods, including Monte Carlo simulation techniques and machine learning applications. Topics include random number generation, variance reduction techniques, importance sampling, and reinforcement learning.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
A solid background in calculus-based probability, multivariable calculus, and linear algebra is recommended.

MA 574 : Portfolio Valuation and Risk Management

Balancing financial risks vs returns by the use of asset diversification is one of the fundamental tasks of quantitative financial management. This course is devoted to the use of mathematical optimization and statistics to allocate assets, to construct and manage portfolios and to measure and manage the resulting risks. The fist part of the course covers Markowitz's mean-variance optimization and efficient frontiers, Sharpe's single index and capital asset pricing models, arbitrage pricing theory, structural and statistical multifactor models, risk allocation and risk budgeting. The second part of the course is devoted to the intertwining of optimization and statistical methodologies in modern portfolio management, including resampled efficiency, robust and Bayesian statistical methods, the Black-Litterman model and robust portfolio optimization.

Department
Mathematical Sciences
Credits 3.0

MA 575 : Market and Credit Risk Models and Management

The objective of the course is to familiarize students with the most important quantitative models and methods used to measure and manage financial risk, with special emphasis on market and credit risk. The course starts with the introduction of metrics of risk such as volatility, value-atrisk and expected shortfall and with the fundamental quantitative techniques used in financial risk evaluation and management. The next section is devoted to market risk including volatility modeling, time series, non-normal heavy tailed phenomena and multivariate notions of codependence such as copulas, correlations and tail-dependence. The final section integrates machine learning techniques, such as deep learning and Monte Carlo methods, to study the valuation of default-contingent claims underlying structural and dynamic models, including credit default swaps, structured credit portfolios, and collateralized debt obligations.

Department
Mathematical Sciences
Credits 3.0
Prerequisites
A solid background in calculus-based probability, multivariable calculus, and linear algebra is recommended.

MA 579 : Financial Programming Workshop

The objective is to elevate the students' computer programming skills to the semi-professional level required in quantitative finance. Participants learn through hands-on experience by working on a structured set of mini projects from computational finance under the guidance of an experienced trainer and the faculty in charge. The programming language used may be C++, MATLAB, R/S, VB or another language widely used in quantitative finance and may alternate from year to year.

Department
Mathematical Sciences
Credits 1.0
Prerequisites
Intermediate scientific programming skills

MA 584/BCB 504 : Statistical Methods in Genetics and Bioinformatics

This course provides students with knowledge and understanding of the applications of statistics in modern genetics and bioinformatics. The course generally covers population genetics, genetic epidemiology, and statistical models in bioinformatics. Specific topics include meiosis modeling, stochastic models for recombination, linkage and association studies (parametric vs. nonparametric models, family-based vs. population-based models) for mapping genes of qualitative and quantitative traits, gene expression data analysis, DNA and protein sequence analysis, and molecular evolution. Statistical approaches include log-likelihood ratio tests, score tests, generalized linear models, EM algorithm, Markov chain Monte Carlo, hidden Markov model, and classification and regression trees. Students may not receive credit for both MA 584 and MA 4603.

Department
Mathematical Sciences
Bioinformatics and Computational Biology
Credits 3.0
Prerequisites
knowledge of probability and statistics at the undergraduate level

MA 590 : Special Topics

Courses on special topics are offered under this number. Contact the Mathematical Sciences Department for current offerings.

Department
Mathematical Sciences
Credits 3.0

MA 595 : Independent Study

Supervised independent study of a topic of mutual interest to the instructor and the student.

Department
Mathematical Sciences
Credits 1.0

MA 596 : Master's Capstone

The Master's Capstone is designed to integrate classroom learning with real-world practice. It can consist of a project, a practicum, a research review report or a research proposal. A written report and a presentation are required.

Department
Mathematical Sciences
Credits 1.0

MA 598 : Professional Master's Project

This project will provide the opportunity to apply and extend the material studied in the coursework to the study of a real-world problem originating in the industry. The project will be a capstone integrating industrial experience with the previously acquired academic knowledge and skills. The topic of the project will come from a problem generated in industry, and could originate from prior internship or industry experience of the student. The student will prepare a written project report and make a presentation before a committee including the faculty advisor, at least one additional WPI faculty member and representatives of a possible industrial sponsor. The advisor of record must be a faculty member of the WPI Mathematical Sciences Department. The student must submit a written project proposal for approval by the Graduate Committee prior to registering for the project.

Department
Mathematical Sciences
Credits 1.0

MA 599 : Thesis

Research study at the masters level.
Department
Mathematical Sciences
Credits 1.0

MA 698 : Ph.D. Project

Ph.D. project work.
Department
Mathematical Sciences
Credits 1.0

MA 699 : Dissertation

Research study at the Ph.D. level.
Department
Mathematical Sciences
Credits 1.0

Mathematics for Educators

MME 518 : Geometrical Concepts

This course focuses primarily on the foundations and applications of Euclidean and non-Euclidean geometries. The rich and diverse nature of the subject also implies the need to explore other topics, for example, chaos and fractals. The course incorporates collaborative learning and the investigation of ideas through group projects. Possible topics include geometrical software and computer graphics, tiling and tessellations, two- and three-dimensional geometry, inversive geometry, graphical representations of functions, model construction, fundamental relationship between algebra and geometry, applications of geometry, geometry transformations and projective geometry, and convexity.

Department
Mathematics for Educators
Credits 3.0